SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4_AT1 — RELATIONSHIP VISUALIZATION LAB

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO4 – Analysis (BL4)	**Assessment:**	Assessment Tool 1 – Relationship Visualization Lab
Weightage:	20% of CIA	**Total Marks:**	25 (5 Questions × 5 Marks)
CO4 Statement:	*Analyze relationships between variables using scatterplots, correlograms, PCA, bubble charts, and network graphs to construct and interpret appropriate visualizations.(BL4)*		
Student Name:	Nivya Sri T	**Reg. No.:**	192424131
Section / Batch:	2024	**Date:**	22-08-26

Code of Conduct

I ________Nivya Sri T_________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ________*Nivya Sri T*___________

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Graph paper or a ruler is recommended for accurate, to-scale plotting.
- Show all supporting calculations (e.g., variance explained, correlation strength, node degree) where relevant.
- Every chart must include a title, correctly labeled axes, and a legend where relevant.

Annexure A – RELATIONSHIP VISUALIZATION LAB

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT1 — Relationship Visualization Lab

CO4 · BT4: Analyze ·

This rubric is used to assess student responses to the CO4-AT1 Relationship Visualization Lab problems, in which students construct and analyze scatterplots, correlograms, PCA outputs, bubble charts, or network graphs from given data.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	**25%**	Demonstrates thorough understanding of the relevant relationship-visualization technique (scatterplot, correlogram, PCA, bubble chart, or network graph) and correctly applies it to the given data.	Good understanding of the relevant technique, with only minor gaps in application.	Basic understanding, but with some misconceptions about the correct technique or how to apply it.	Poor understanding of core concepts; applies a technique fundamentally unsuited to the data.
Explanation	**25%**	Provides detailed, well-reasoned analysis of the constructed visualization, explicitly tied to the specific values in the dataset.	Provides clear analysis with adequate reasoning tied to the data.	Analysis is present but generic or only loosely connected to the specific dataset given.	Little or no analytical reasoning provided; answers restate the data without interpretation.
Application	**20%**	Effectively applies construction and analysis techniques to produce a complete, accurate visualization correctly matched to the data.	Applies techniques to produce a correct visualization, with only minor errors.	Limited application; the visualization partially represents the data but has notable errors.	Fails to apply appropriate techniques; the visualization does not correctly represent the data.
Organization	**15%**	The response is clearly structured, with construction and analysis presented in a logical sequence with coherent overall flow.	The response is organized, with only minor issues in flow or clarity.	Basic structure; construction and analysis are present but the response lacks clarity or sequence.	Poorly organized; the response is incoherent or key parts are left unaddressed.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Accuracy	**10%**	All parts of the answer — construction, calculations, and analysis — are completely accurate and well-suited to the given data.	Mostly accurate, with only minor omissions or a small error.	Partially correct; some elements are missing, mismatched to the data, or incorrect.	Largely incorrect or inappropriate, with major gaps across multiple parts.

Scoring Guide

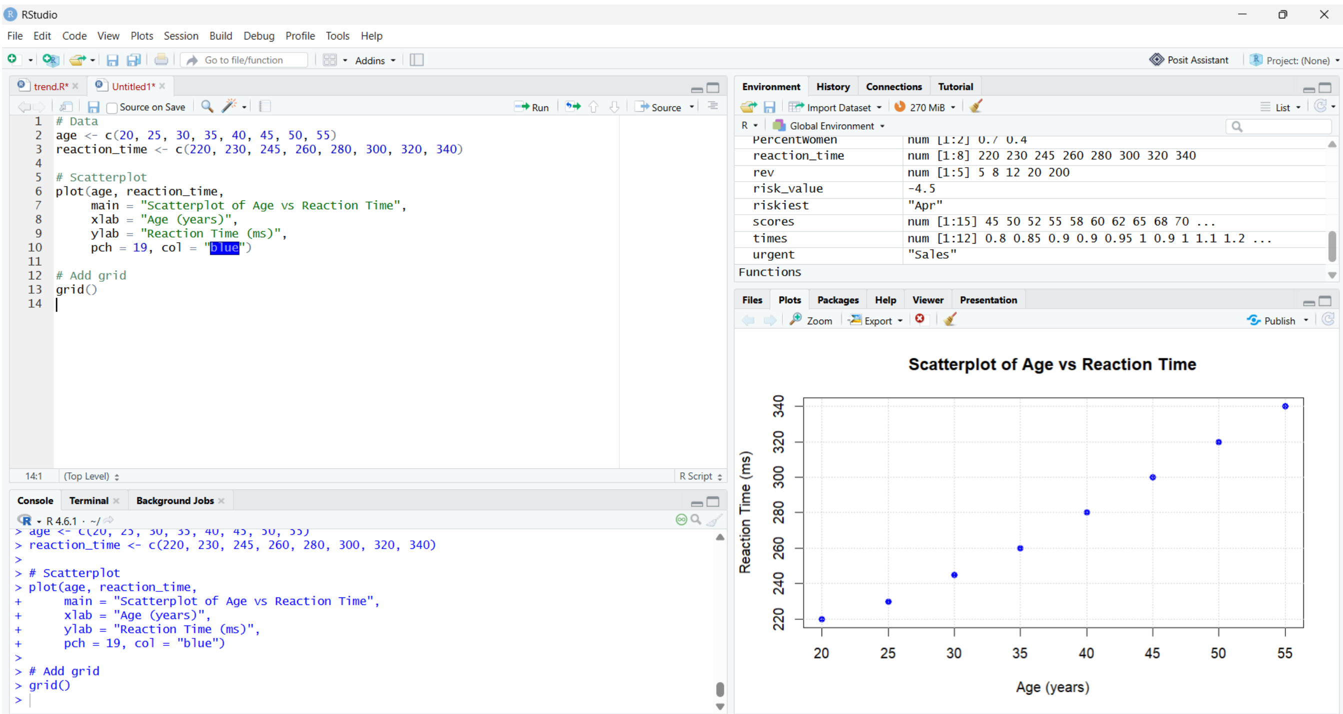
For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

PROBLEM 3: Scatterplot of Age vs Reaction Time

Description of the Relationship

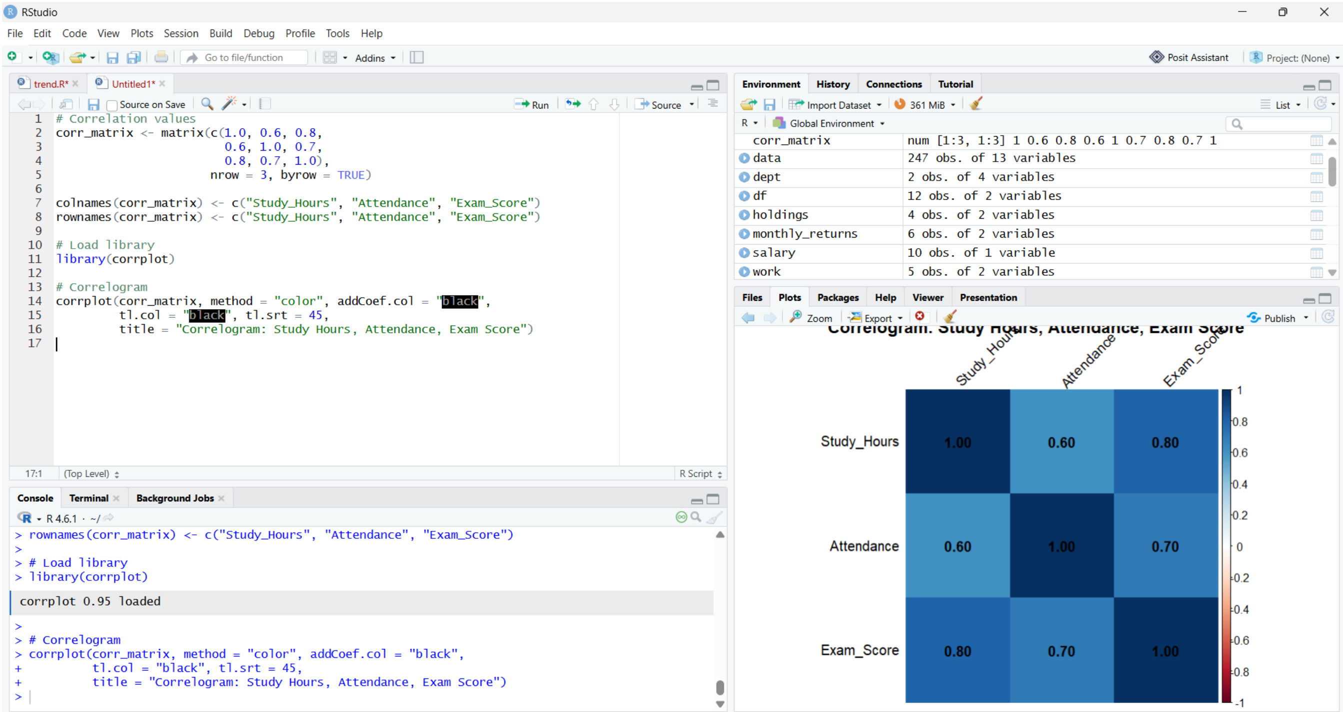
- As age increases, reaction time also increases.
- The points form a clear upward trend, showing a positive linear relationship.
- Older participants have slower reaction times.



PROBLEM 7: Correlogram

Interpretation

- SH–Score (0.8) is the strongest relationship → students who study more tend to score higher.
- Att–Score (0.7) is also strong → consistent attendance improves exam performance.
- SH–Att (0.6) is moderate → students who study more often also attend more, but the link is weaker than the others.



PROBLEM 11: Mean of Each Variable

Let's compute the means using your dataset:

Heights (cm): 150, 155, 160, 165, 170, 175 Weights (kg): 48, 52, 58, 62, 68, 72

Mean Height

$$\text{Mean Height} = \frac{150 + 155 + 160 + 165 + 170 + 175}{6} = \frac{975}{6} = 162.5 \text{ cm}$$

Mean Weight

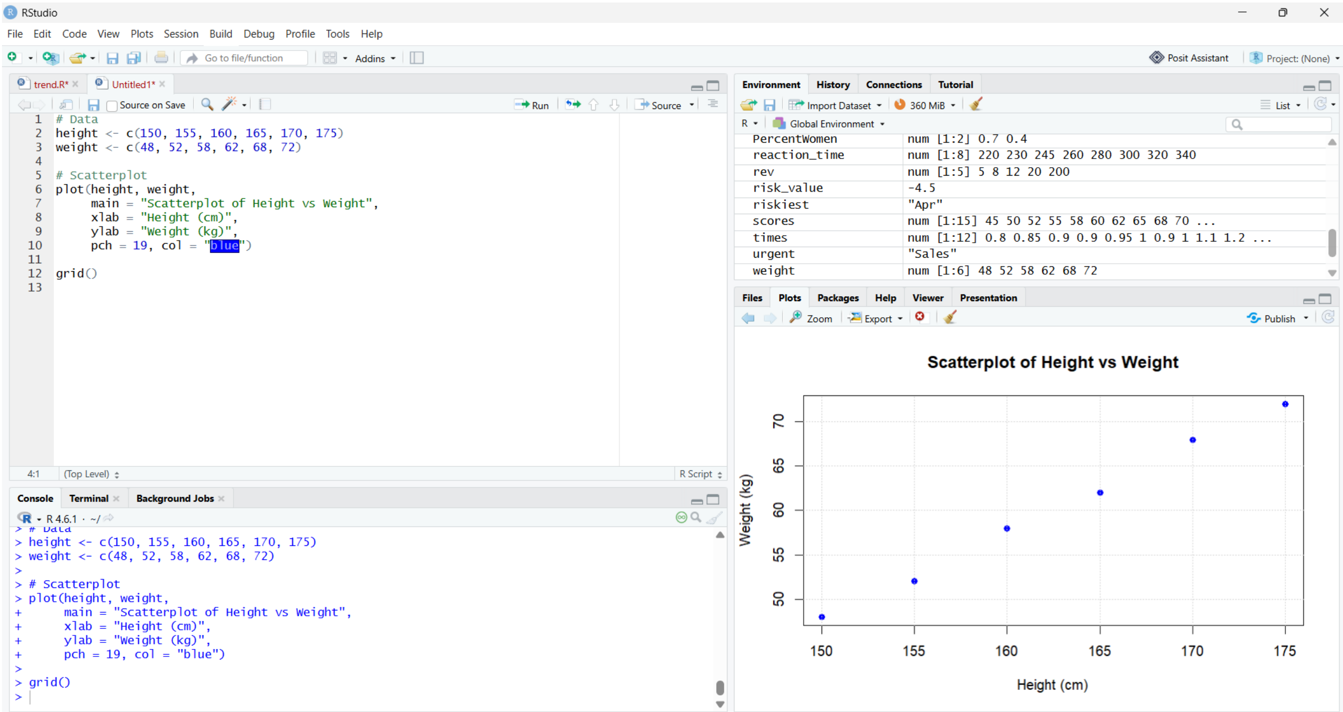
$$\text{Mean Weight} = \frac{48 + 52 + 58 + 62 + 68 + 72}{6} = \frac{360}{6} = 60 \text{ kg}$$

What PC1 Represents in This Dataset

With only two variables (Height and Weight), Principal Component 1 (PC1) captures the main direction of variation in the data.

Interpretation of PC1

- Height and weight increase together in your dataset.
- PC1 will represent a combined growth factor — essentially a "body size" dimension.
- Individuals with higher height and weight will have higher PC1 scores.
- PC1 explains most of the variance, because both variables are strongly positively related.



PROBLEM 20: Bubble Chart for Team Performance

We evaluate all three variables:

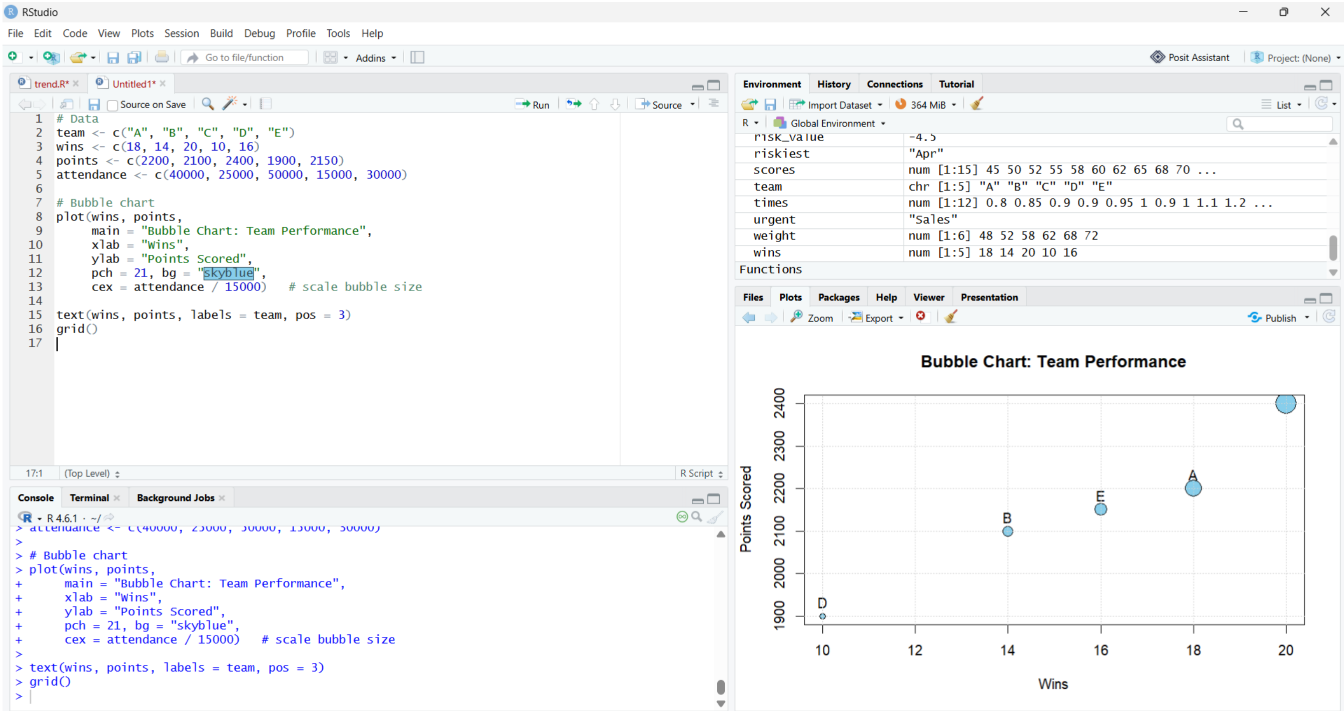
1. Wins
   - Highest: Team C (20 wins)
2. Points Scored
   - Highest: Team C (2400 points)
3. Fan Attendance (Bubble Size)
   - Highest: Team C (50,000 fans)

⭐ Best Overall Team: TEAM C

Team C dominates every metric:

- Most wins
- Most points
- Largest fan base

So Team C clearly has the strongest overall performance.



PROBLEM 24: Collaboration Network Graph

Look at the collaboration links:

Group 1

- R1–R2
- R1–R3
- R2–R3

These three form a fully connected triangle → one collaboration cluster.

Group 2

- R4–R5
- R5–R6

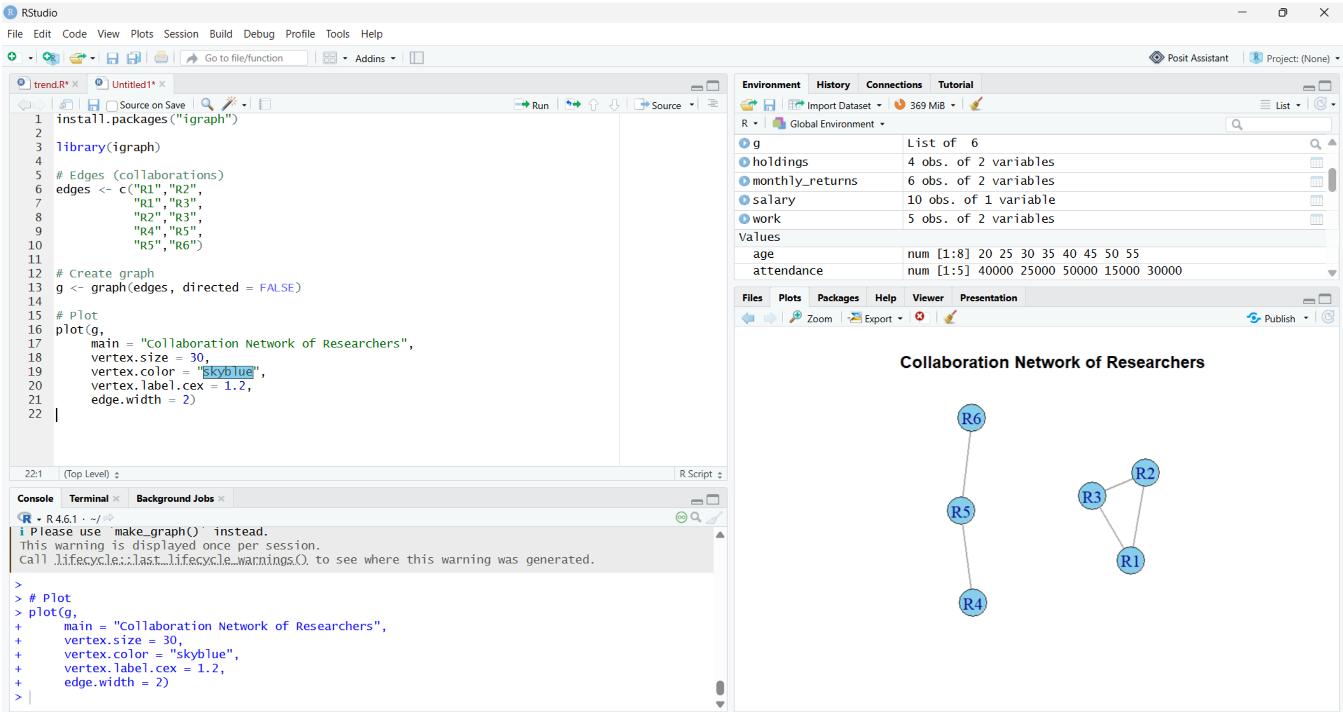
These three form a chain-like cluster → second collaboration group.

⭐ Final Answer

The network forms two separate groups, not one connected group.

- Cluster 1: R1, R2, R3
- Cluster 2: R4, R5, R6

There is no link between the two clusters, so the network is disconnected.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Posit Assistant
Project: (None)
trend.R*
Untitled1*
Source on Save
Run
Source
install.packages("igraph")
library(igraph)
# Edges (collaborations)
edges <- c("R1","R2",
"R1","R3",
"R2","R3",
"R4","R5",
"R5","R6")
# Create graph
g <- graph(edges, directed = FALSE)
# Plot
plot(g,
main = "Collaboration Network of Researchers",
vertex.size = 30,
vertex.color = "skyblue",
vertex.label.cex = 1.2,
edge.width = 2)
22:1
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · ~/
i Please use `make_graph()` instead.
This warning is displayed once per session.
Call lifecycle::last_lifecycle_warnings() to see where this warning was generated.
> # Plot
> plot(g,
+ main = "Collaboration Network of Researchers",
+ vertex.size = 30,
+ vertex.color = "skyblue",
+ vertex.label.cex = 1.2,
+ edge.width = 2)
Environment
History
Connections
Tutorial
Import Dataset
369 MiB
List
Global Environment
g
List of 6
holdings
4 obs. of 2 variables
monthly_returns
6 obs. of 2 variables
salary
10 obs. of 1 variable
work
5 obs. of 2 variables
Values
age
num [1:8] 20 25 30 35 40 45 50 55
attendance
num [1:5] 40000 25000 50000 15000 30000
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Collaboration Network of Researchers
R6
R5
R4
R2
R3
R1